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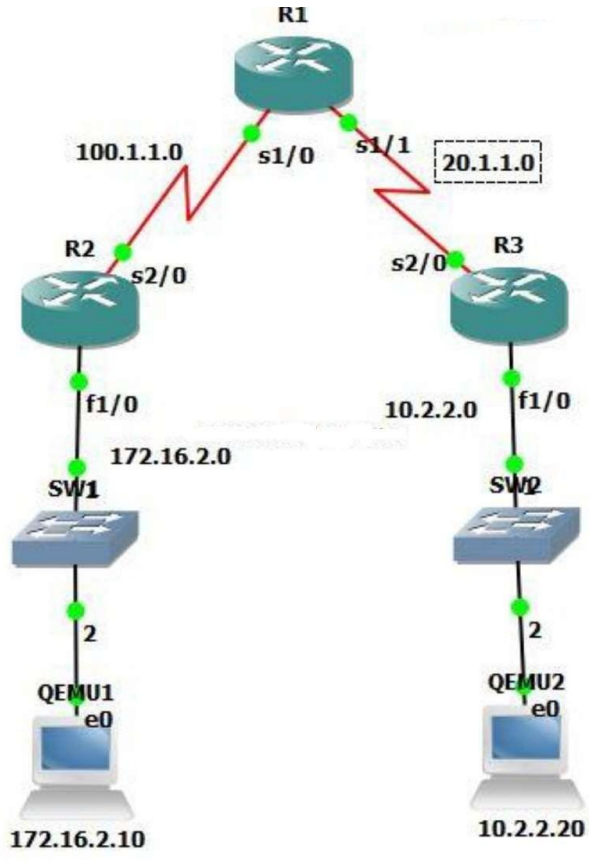
Assignment 6: Solution
Routing using RIP Version 2 (Cisco Packet Tracer)

Objective

To configure and verify routing using RIP Version 2 protocol in a network.

Problem Statement

Design and implement the given network topology in Cisco Packet Tracer with three routers (R1, R2, R3), two LANs, and PCs using the provided IP addressing scheme.



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Network Details

Device	Interface	IP Address	Subnet Mask
R1	S0/0/0	100.1.1.2	255.255.255.0
R1	S0/0/1	20.1.1.1	255.255.255.0
R2	S0/0/0	100.1.1.1	255.255.255.0
R2	G0/0	172.16.2.1	255.255.0.0
R3	S0/0/1	20.1.1.2	255.255.255.0
R3	G0/0	10.2.2.1	255.255.255.0
PC1	NIC	172.16.2.10	255.255.0.0
PC2	NIC	10.2.2.20	255.255.255.0

Tasks

- Create the network topology in Cisco Packet Tracer.
- Configure IP addresses on all router interfaces and PCs.
- Enable RIP Version 2 routing on all routers.
- Advertise all directly connected networks using RIP.
- Disable auto-summary in RIP configuration.
- Verify routing tables using 'show ip route'.
- Test connectivity by pinging PC2 from PC1 and between routers.

Router Configurations

R1 Configuration:

```
enable
conf t
interface s0/0/0
ip address 100.1.1.2 255.255.255.0
clock rate 64000
no shutdown
interface s0/0/1
ip address 20.1.1.1 255.255.255.0
no shutdown
router rip
version 2
no auto-summary
```

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```
network 100.1.1.0
network 20.1.1.0
end
```

R2 Configuration:

```
enable
conf t
interface s0/0/0
ip address 100.1.1.1 255.255.255.0
no shutdown
interface g0/0
ip address 172.16.2.1 255.255.0.0
no shutdown
router rip
version 2
no auto-summary
network 100.1.1.0
network 172.16.0.0
end
```

R3 Configuration:

```
enable
conf t
interface s0/0/1
ip address 20.1.1.2 255.255.255.0
no shutdown
interface g0/0
ip address 10.2.2.1 255.255.255.0
no shutdown
router rip
version 2
no auto-summary
network 20.1.1.0
network 10.2.2.0
end
```

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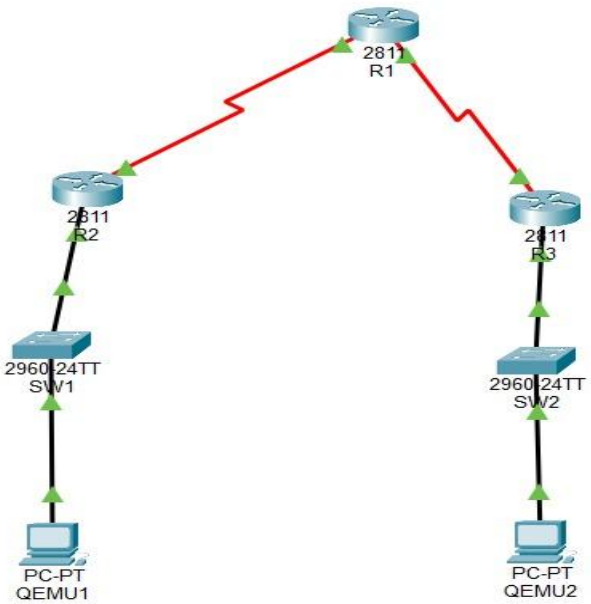
Solution